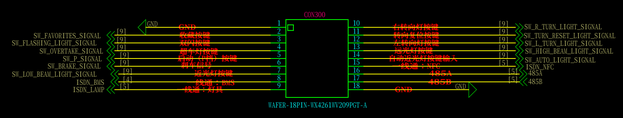
V2.0

1.删除:喇叭按键输入&位置灯按键输入&加热按键输入&近光灯按键输入检测
2、新增:超车灯按键输入&自动大灯按键输入&近光常开按键输入&收藏键按键输入检测
3、删除:近光灯控制&远光灯控制&左补光灯&右补光灯&刹车灯&紧急刹车灯&前后位置灯删除喇叭控制
4、新增灯具控制:牌照灯&座桶灯&LOGO灯)&一线通控制（牌照灯(正控，12V)座桶灯(正控，12V)LOGO灯(正控，12V)-线通控制(5V,收发)）
5、左右刹车信号检测合并为一路
6、删除一路12V DCDC输入 ，改为加热DC输入。
7、删除:ACC1仪表控制、删除雷达正极控制
8、新版本龙头锁/坐垫锁驱动电路变更
9、新增NFC 电源5V，通讯方式:一线通
10、新增:充电指示灯1&蓝牙指示灯&仪表复位
11、新增RS485 原预留串口改为485通讯
12、删除NTC-&坐垫温感-
13、新增充电输入检测 新增BMS一线通(5V,收发)

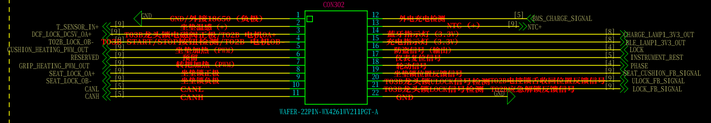
1	项目	内容	备注
2	组合开关	1.删除：喇叭按键输入检测&位置灯按键输入&加热按键输入&近光灯按键输入 2.新增：超车灯按键输入&自动大灯按键输入&近光常开按键输入&收藏键按键输入	按键输入，低有效
3	刹车	左右刹车输入合并为一路刹车输入	
4	灯具	1.删除：近光灯控制&远光灯控制&左补光灯&右补光灯&刹车灯&紧急刹车灯&前后位置灯 2.新增灯具控制：牌照灯&座桶灯&LOGO灯)&一线通控制	牌照灯(正控，12V) 座桶灯(正控，12V) LOGO灯(正控，12V) 一线通控制(5V,收发)
5	喇叭	删除喇叭控制	
6	DCDC	1.删除一路DCDC输入	原CON20-PIN9
7	ACC	删除：ACC1仪表控制	
8	雷达	删除雷达正极控制	
9	坐垫锁	1.移除开坐垫锁输入检测 2.匹配坐垫锁型号：M179Z(5V，带反馈信号)	4PIN：电机正极/电机负极/电机反馈信号/GND
10	龙头锁	匹配新的龙头锁(雷牌)，兼容中立4代龙头锁	
11	坐垫传感器	新增坐垫传感器接口，低有效	
12	备用电池	使用18650 2000mah电池	支持主电丢失后开坐垫锁功能
13	充电检测	新增充电输入检测	低有效
14	NFC	新增NFC	电源5V，通讯方式:一线通
15	仪表	新增:充电指示灯&蓝牙指示灯&仪表复位	指示灯:3.3V 仪表复位:低有效
16	RS485	新增RS485	原预留串口改为485通讯
17	数据存储	存储音频的FLASH改为8MB	
18	其他	1.删除NTC-&坐垫温感- 2.新增一线通(5V,收发)	NTC-：原CON22-PIN3 坐垫温感-：原CON22-PIN6

18PIN线束接口

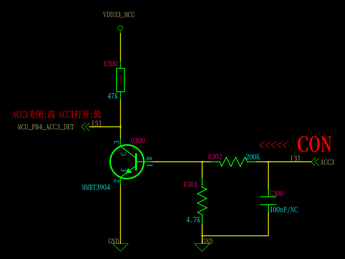


20PIN线束接口

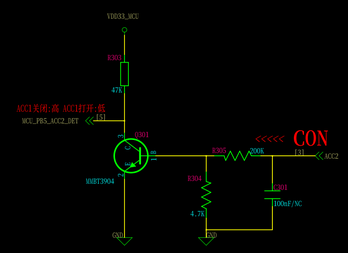
22PIN线束接口



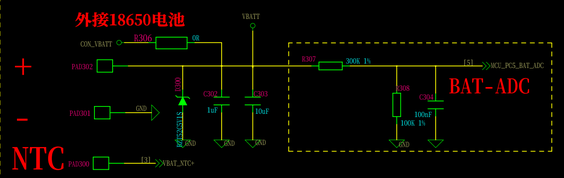
ACC3检测



ACC2检测

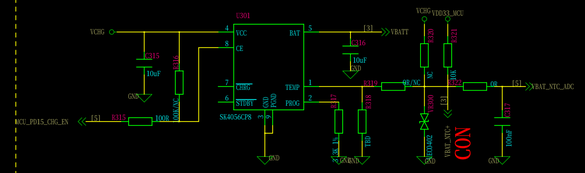


BATTERY



Linear Charger RM--1100mAH

CHARGE:广芯1100/R315=363mA；友台1200/R315=333ma

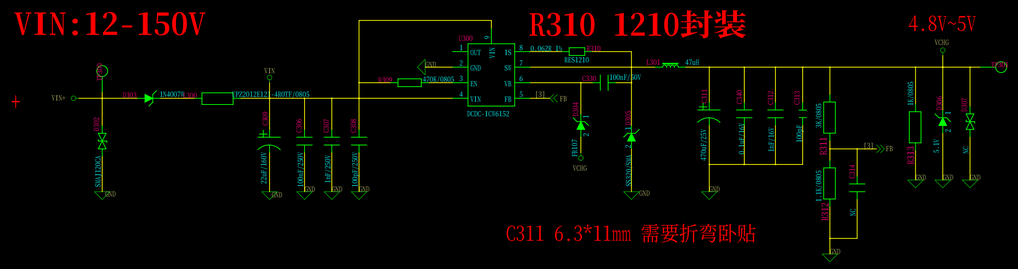


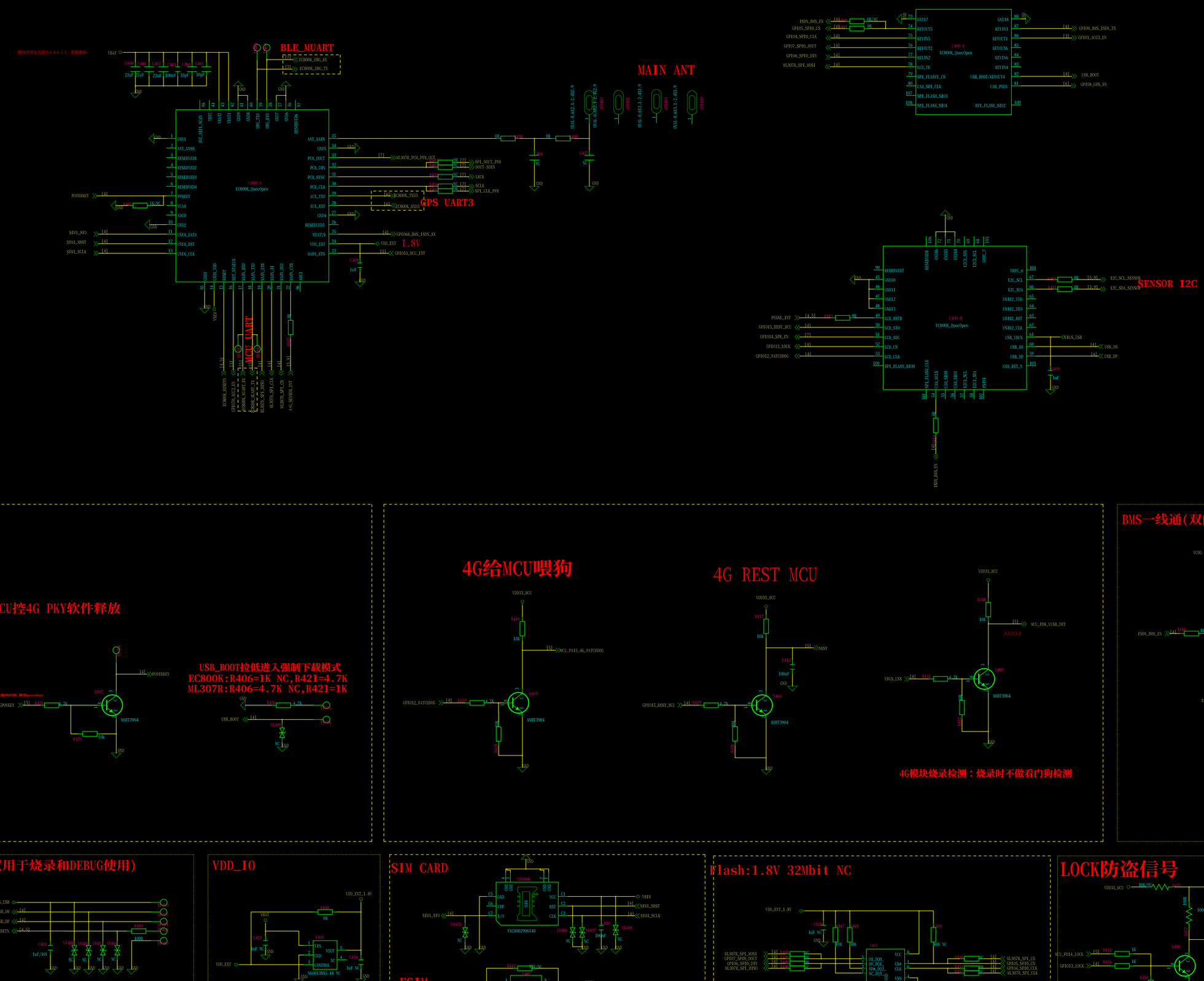
VIN:12-150V

R310 1210封装

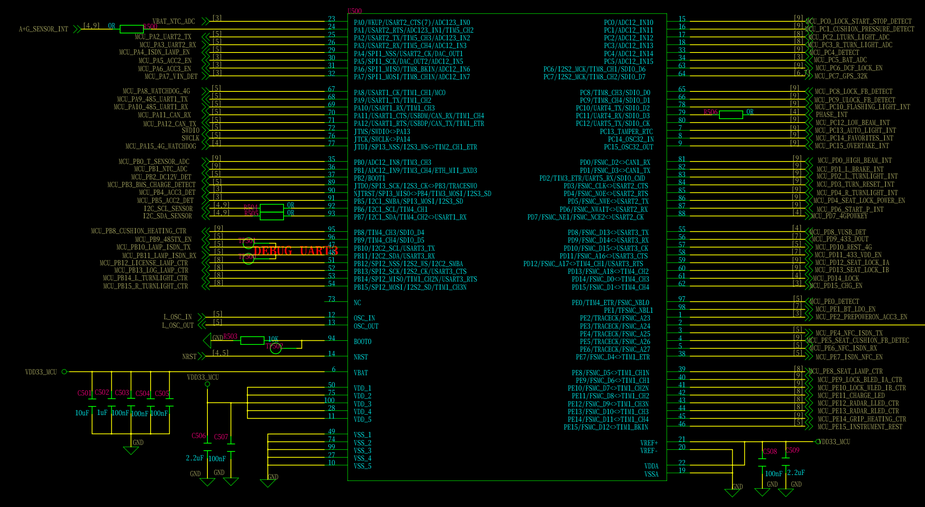
2A左右

4.8V~5V

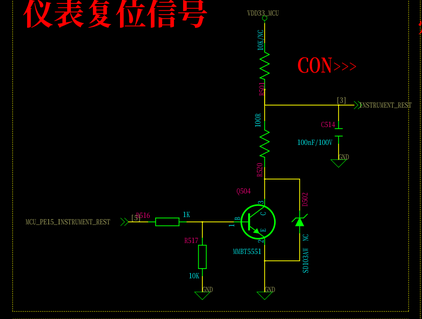



BLE_MUART
MAIN ANT
GPS UART3
1.8V
MCU UART
SENSOR I2C
U400-A
EC800K_Quec0pen
U400-B
U400-C
U400-D
ANT_MAIN
PCI_DOUT
PCI_DIN
PCI_SYNC
PCI_CLK
AUX_TXD
AUX_RXD
STATUS
VDD_EXT
MAIN_RTS
POWERKEY
USB_BOOT
LCD_RSTB
LCD_SI0
LCD_SDC
LCD_CS
LCD_CLK
SPI_FLASH_SI100
USB_DM
USB_DP
CAN_RST_N
USB_VBUS
I2C_SCL
I2C_SDA
GPIO68_BHS_ISDN_RX
GPIO100_BHS_ISDN_TX
GPIO101_ACC1_EN
GPIO53_MCU_INT
GPIO15_REST_MCU
GPIO14_SPK_EN
GPIO13_LOCK
GPIO12_WATCHDOG
PHASE_INT
SIM1_SIO
SIM1_SRST
SIM1_SCLK
SPI_DOUT_PWM
DOUT-SO1N
LRCK
SCLK
SPI_CLK_PWM
VDD_EXT
VBAT
VCHG
VDD33_MCU
MCU_PA15_4G_WATCHDOG
MCU_PD8_VUSB_DET
MCU_PD7_4GPOWKEY
VBUS_USB
USB_DM
USB_DP
LOCK
PHASE
PHASE_INT
CON>>>
0~72V
MAIN ANT
BMS一线通(双向) 【4G模块】
总线 <<<<>>>
>>>>>
<<<<<一线通发
一线通收 >>>>>
4G给MCU喂狗
4G REST MCU
4G模块烧录检测：烧录时不做看门狗检测
MCU控4G PKY软件释放
USB_BOOT拉低进入强制下载模式
EC800K:R406=1K NC,R421=4.7K
ML307R:R406=4.7K NC,R421=1K
USB(仅用于烧录和DEBUG使用)
VDD_IO
SIM CARD
ESIM
Flash:1.8V 32Mbit NC
LOCK防盗信号
3.3V低有效
相位轮动

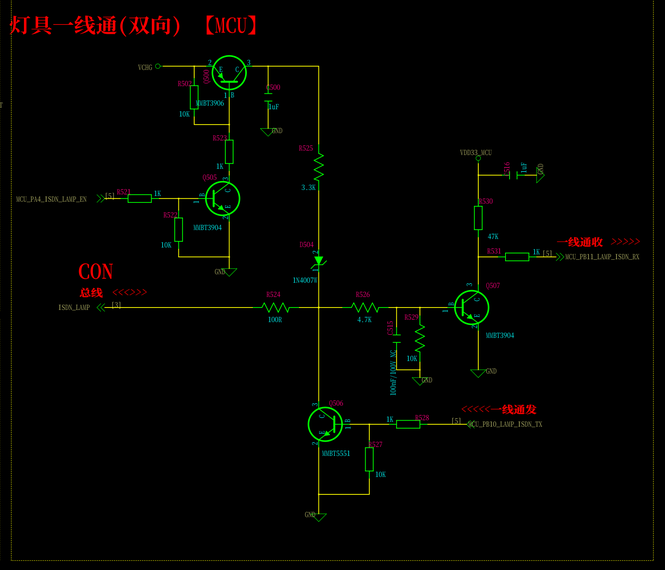
MCU：APM32E103VCT6



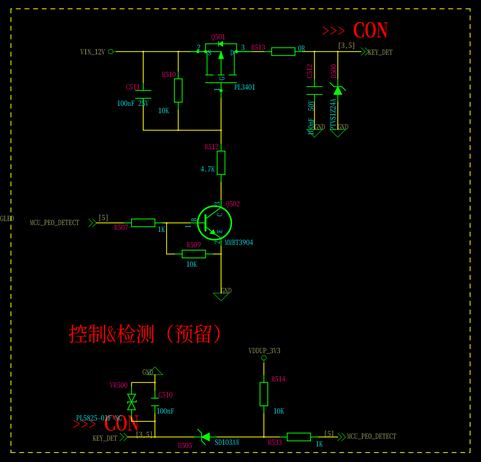
仪表复位信号



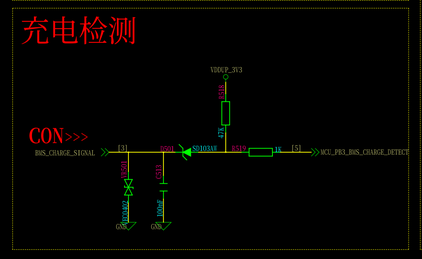
灯具一线通(双向) 【MCU】



>>> CON



充电检测

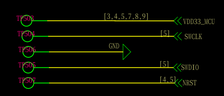


控制&检测（预留）

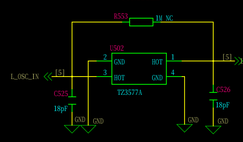
MCU 3.3V

上拉电平预留

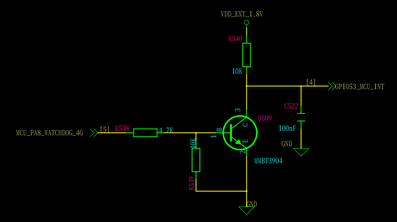
MCU烧录排插



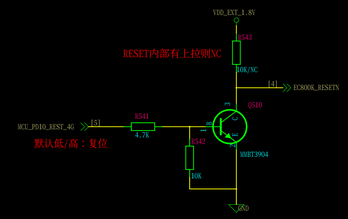
8MHZ



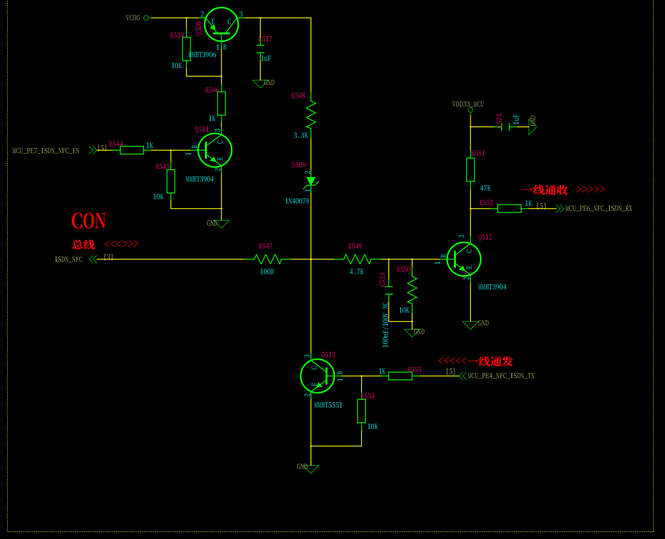
MCU给4G喂狗



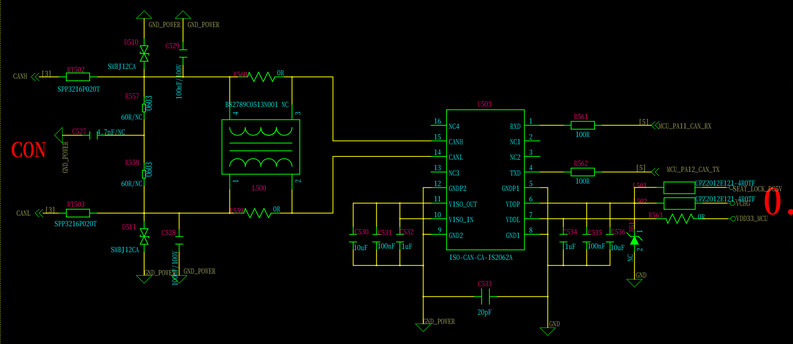
MCU REST 4G



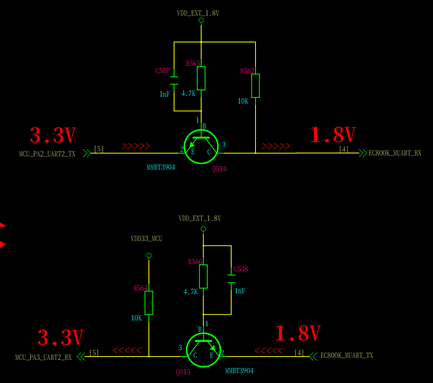
NFC一线通(双向) 【MCU】



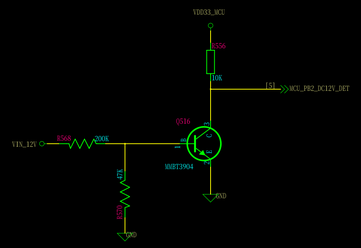
隔离CAN



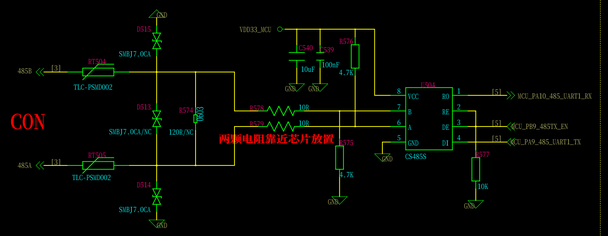
MCU--CAT1 LEVEL SHIFT



预留12V检测

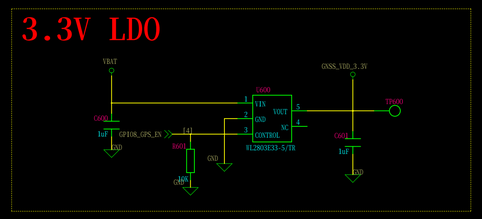


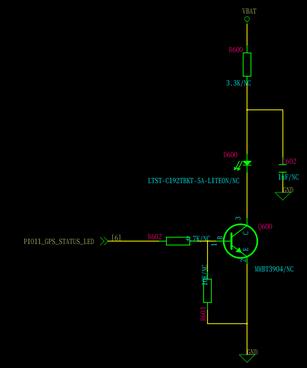
RS485

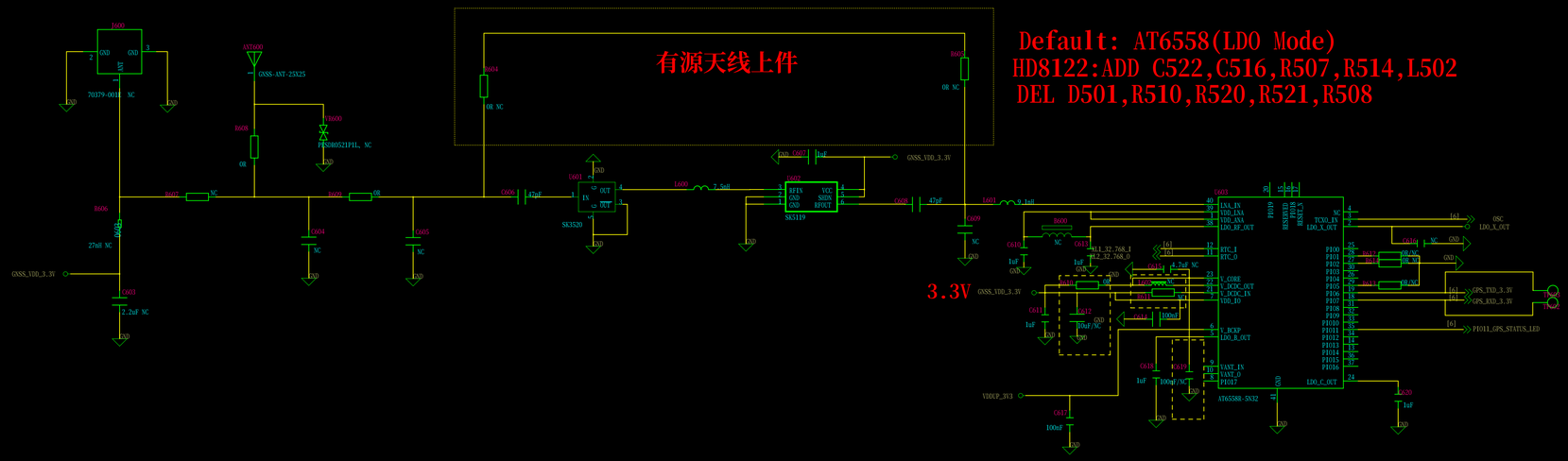


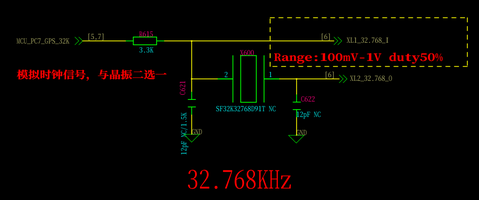
CAN 总线通常有两种状态：显性电平和隐性电平。显性时，CAN_H 和 CAN_L 之间的电压差是大约 2.5V（比如 CAN_H=3.5V，CAN_L=1.0V），隐性时两者都是 2.5V 左右，电压差为 0V。

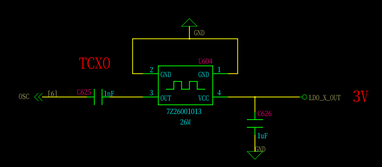
GPS:GREEN

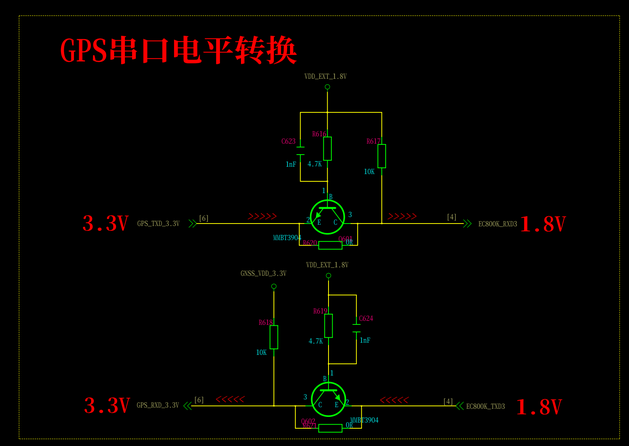

3.3V LDO
VBAT
GNSS_VDD_3.3V
U600
VIN
VOUT
GND
NC
CONTROL
XL2803E33-5/TR
GP108_GPS_EN
R601
10k
C602
1uF
C601
1uF
TP600


VBAT
R600
3.3K/NC
D600
LTST-C192TBKT-5A-LITEON/NC
C602
1uF/NC
Q600
MMBT3904/NC
PI011_GPS_STATUS_LED
R602
2K/NC
R603
10K/NC

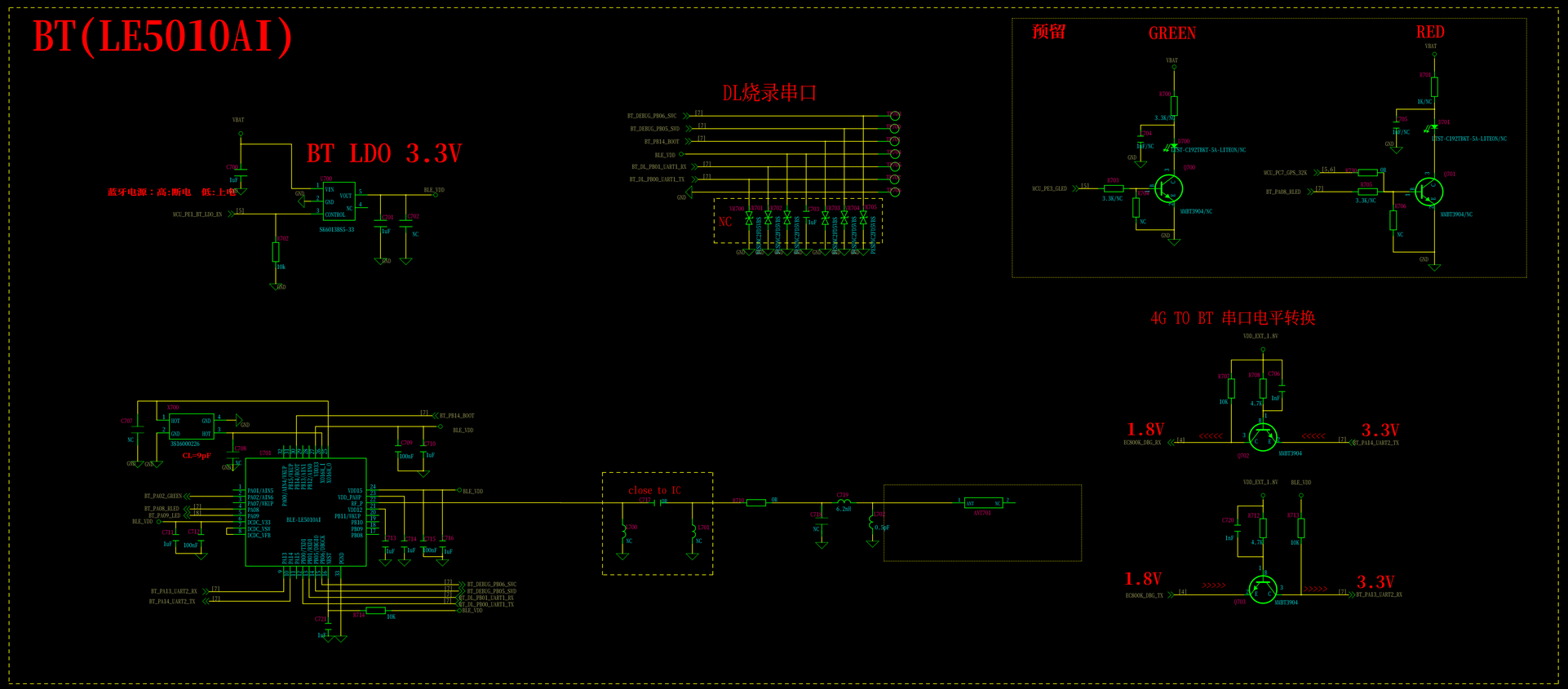

有源天线上件
Default: AT6558(LDO Mode)
HD8122:ADD C522,C516,R507,R514,L502
DEL D501,R510,R520,R521,R508
J600
70179-0014 NC
ANT600
GNSS-ANT-25X25
R608
0R
TVS600
ESD052F1P1L NC
R606
27nH NC
GNSS_VDD_3.3V
C603
2.2uF NC
R602
NC
R604
0R NC
R605
0R
C604
NC
C605
NC
C606
47pF
U601
SK3520
L600
2.5nH
U602
SK5119
C607
1uF
C608
47pF
L601
0.1nH
R603
0R NC
C609
NC
3.3V
C610
1uF
C611
1uF
C612
10uF/NC
R600
NC
C613
32.768_I
32.768_O
C614
1.2uF NC
C615
100nF
C616
1uF
C617
100nF
C618
1uF
C619
100nF/NC
VDDCP_3V3
C603
LNA_IN
VDD_LNA
VDD_ANA
LDO_RF_OUT
RTC_I
RTC_O
V_CORE
V_DCDC_OUT
V_DCDC_IN
VDD_IO
V_BCKP
LDO_B_OUT
VANT_IN
VANT_O
PI017
AT6558R-5X32
PI019
HRSSTRED
PI018
RESET_5
NC
TCXO_IN
LDO_X_OUT
PI000
PI001
PI002
PI003
PI004
PI005
PI006
PI007
PI008
PI009
PI010
PI011
PI012
PI013
PI014
PI015
PI016
LDO_C_OUT
C620
1uF
OSC
LDO_X_OUT
C616
NC
C612
0R NC
C614
0R NC
GPS_TXD_3.3V
GPS_RXD_3.3V
PI011_GPS_STATUS_LED


divided to 0.9V
WCU_PC7_GPS_32K
R612
3.3K
Range:100mV-1V duty50%
模拟时钟信号，与晶振二选一
Y600
SF32K32768094T NC
C621
12pF NC
C622
12pF NC
XL1_32.768_I
XL2_32.768_O
32.768KHz

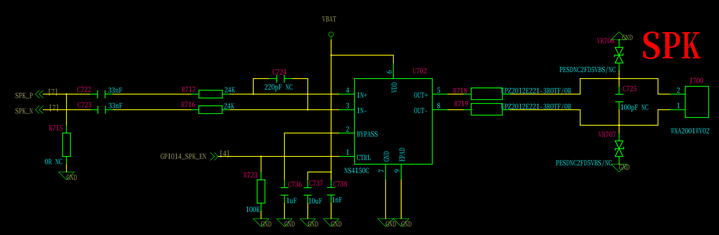

TCXO
U604
OSC
C625
1uF
7Z26001033
26M
C626
1uF
LDO_X_OUT
3V


GPS串口电平转换
VDD_EXT_1.8V
R611
4.7k
R613
10k
C623
1uF
3.3V
GPS_TXD_3.3V
MMBT3904
FC8000K_RX033
1.8V
GNSS_VDD_3.3V
R612
10k
C624
1uF
3.3V
GPS_RXD_3.3V
FC8000K_TX033
1.8V

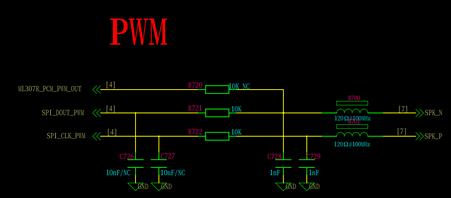
BT(LE5010AI)

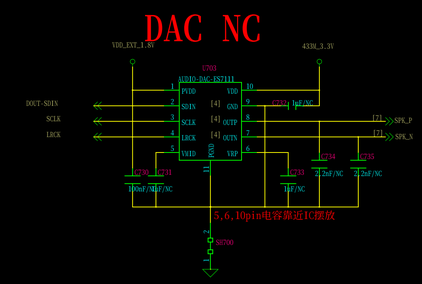


CLASS D
(4ohm 2W)

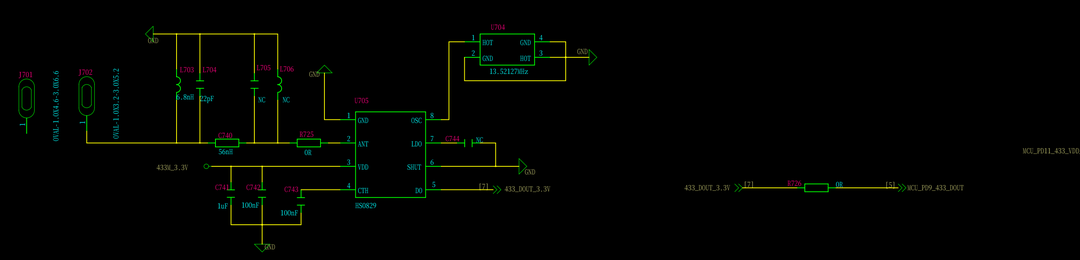


二选一上件



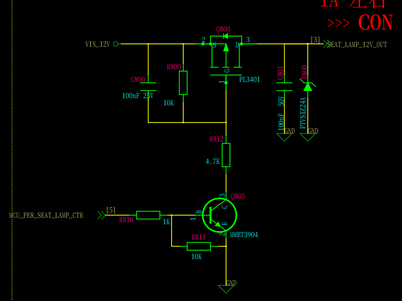


RF接收器433MHZ



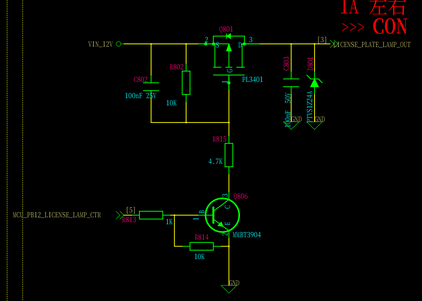
坐桶灯控制

1A 左右
>>> CON



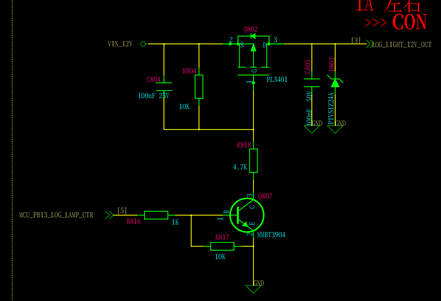
牌照灯控制

1A 左右
>>> CON



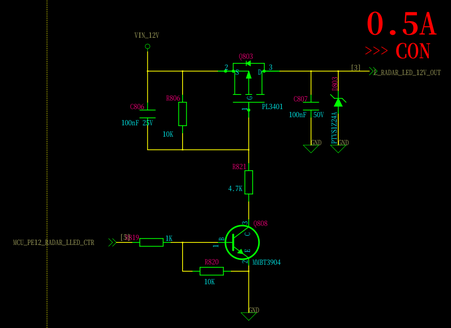
LOG灯控制

1A 左右
>>> CON



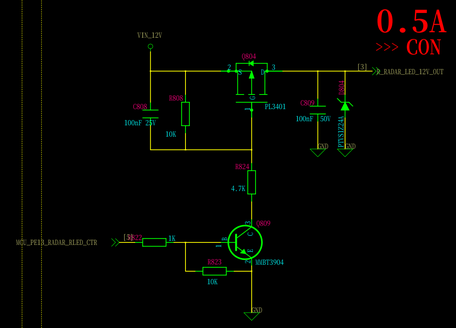
雷达左后视镜补光灯控制

0.5A
>>> CON



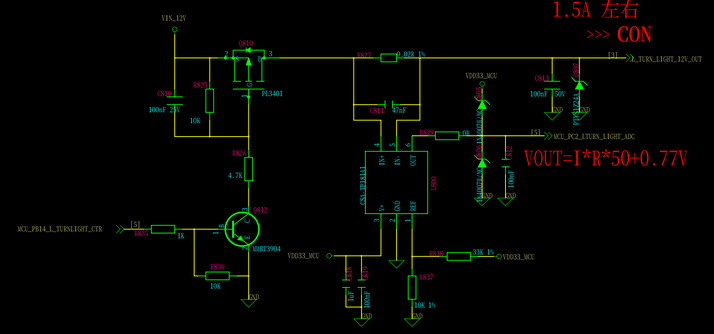
雷达右后视镜补光灯控制

0.5A
>>> CON



左转向灯控制

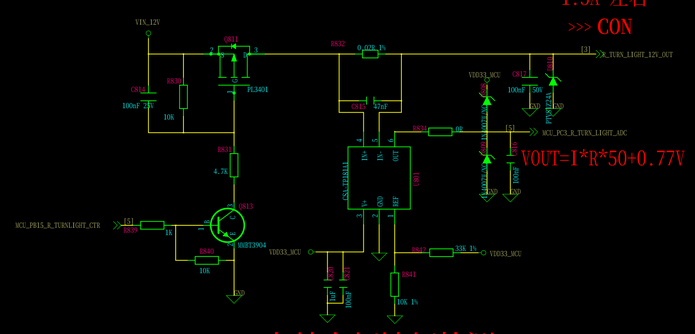
1.5A 左右
>>> CON



左转向灯缺灯检测

右转向灯控制

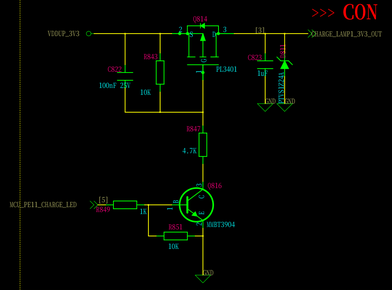
1.5A 左右
>>> CON



右转向灯缺灯检测

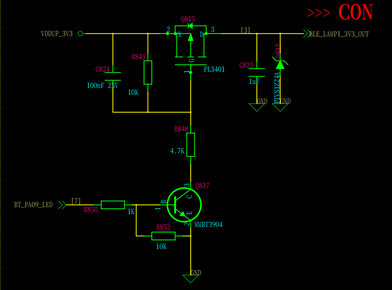
充电状态指示灯1

0.5A
>>> CON

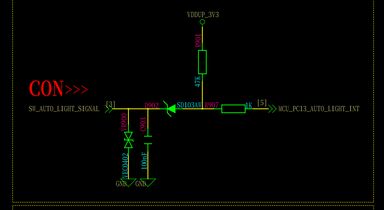


蓝牙状态指示灯控制

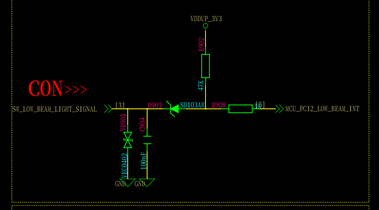
0.5A
>>> CON



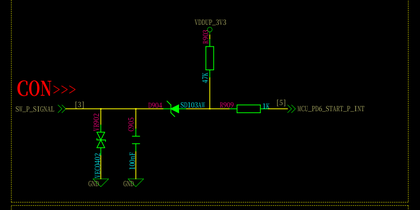
自动近光灯按键检测



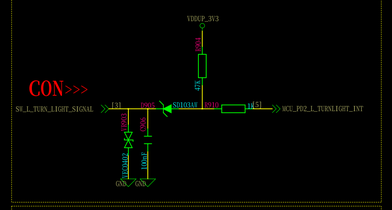
近光灯常开按键检测



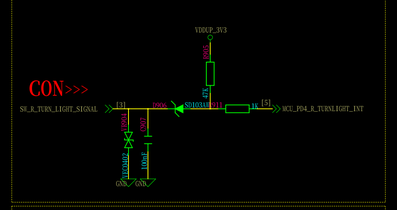
启动按键检测



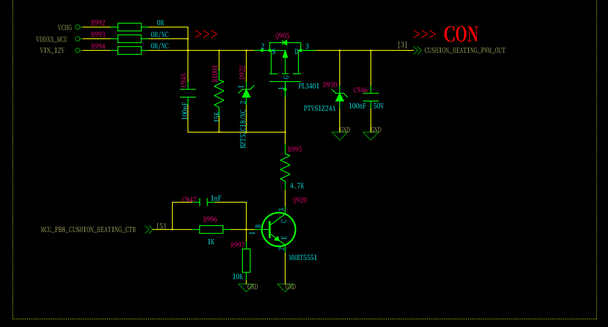
左转向灯按键检测



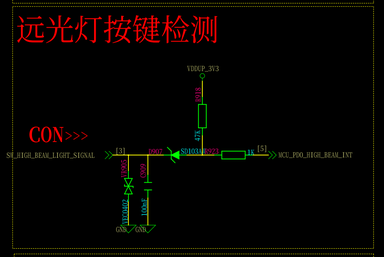
右转向灯按键检测



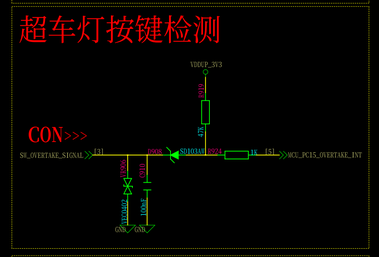
座垫加热控制 PWM



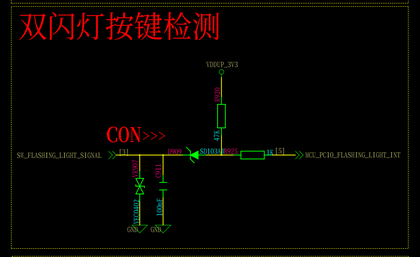
远光灯按键检测



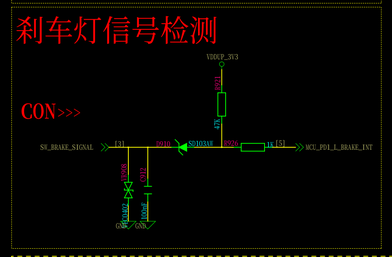
超车灯按键检测



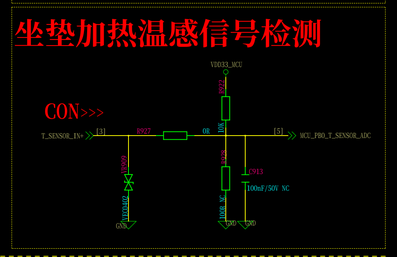
双闪灯按键检测



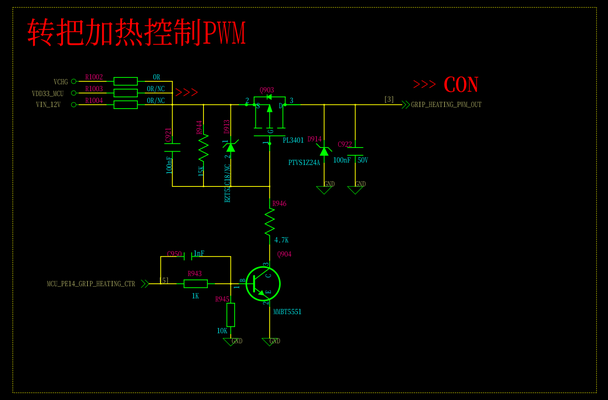
刹车灯信号检测



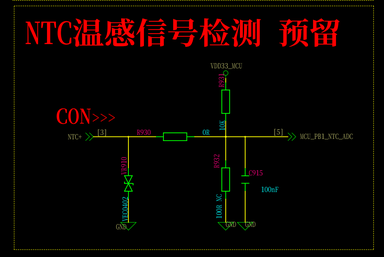
坐垫加热温感信号检测



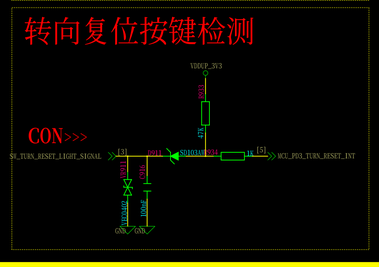
转把加热控制PWM



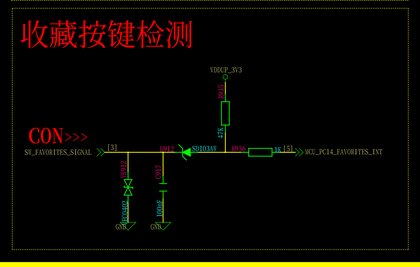
NTC温感信号检测 预留



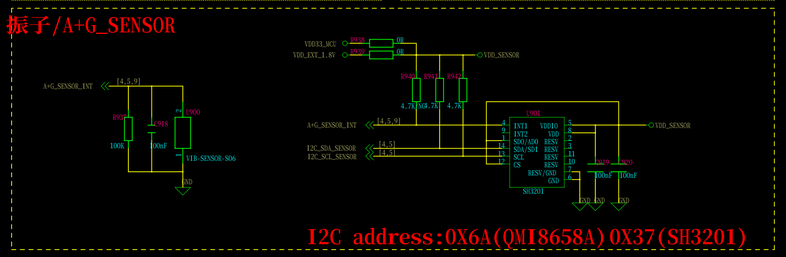
转向复位按键检测



收藏按键检测



振子/A+G_SENSOR

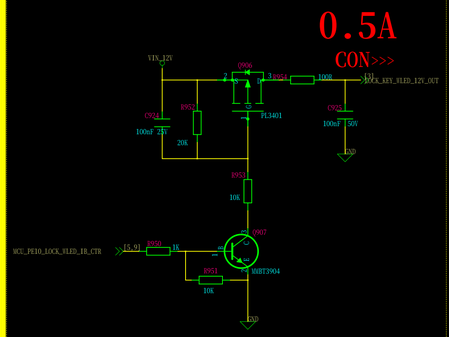


I2C address:0X6A(QMI8658A)0X37(SH3201)

T03B龙头锁：

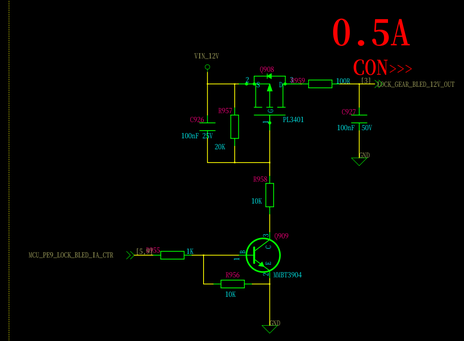
T03B开关组件LED指示灯控制

0.5A

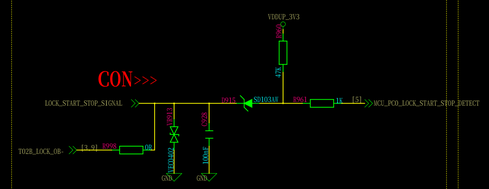


T03B档位盖LED指示灯控制-蓝色

0.5A

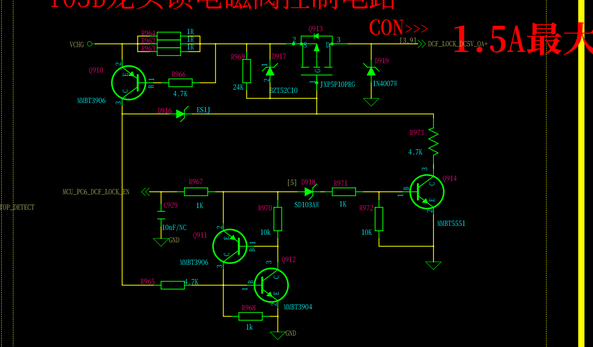


T03B START/STOP按钮检测



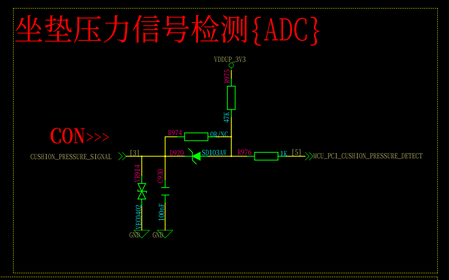
T03B龙头锁电磁阀控制电路

1.5A最大



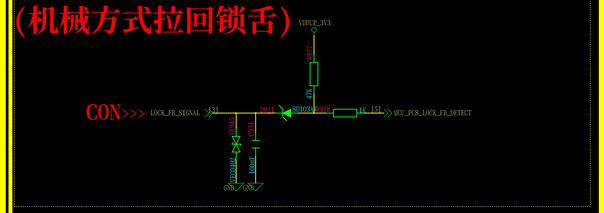
4-5V@0.3A（max）@16.6Ω

坐垫压力信号检测{ADC}

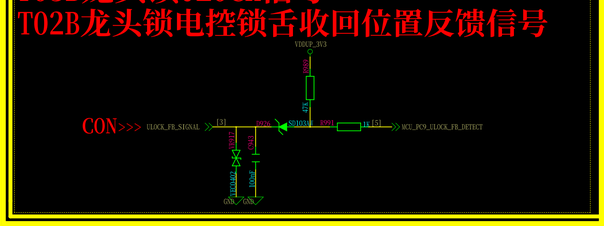


T02B龙头锁：

T03B龙头锁LOCK信号 T02B龙头锁应急解锁反馈信号（机械方式拉回锁舌）

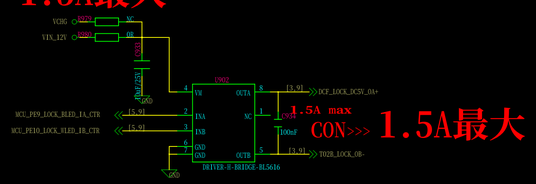


T03B龙头锁ULOCK信号 T02B龙头锁电控锁舌收回位置反馈信号



T02B龙头锁 （LTS04 龙头锁）

1.5A最大

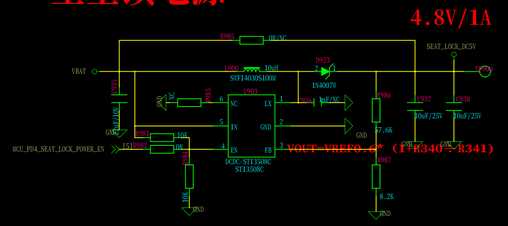


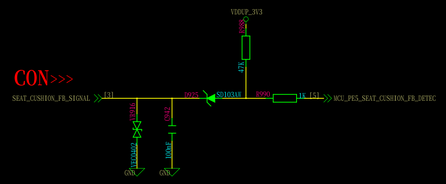
1.5A最大

DC2.0-6.0V@《750mA@马达电阻5.5±12%Ω
堵转电流最大值≤1480mA（max）

坐垫锁电源

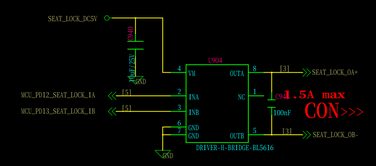
4.8V/1A





坐垫锁状态检测，开为低 关为高

坐垫锁控 （M179Z电动座垫锁）



1.5A最大

坐垫锁DC3.0-7.0V@330mA@马达电阻5±15%Ω
坐垫锁堵转电流最大值1.5A（max）
堵转时间不能超过3s